Appendix 1. Generating an MID relative risk threshold

Core GRADE users will, when outcomes are binary, typically judge inconsistency by examining forest plots presenting relative risks and considering the similarity of point estimates and the extent to which confidence intervals overlap. When the threshold is the MID, however, it is absolute rather than relative differences that determine whether inconsistency is problematic. Making the judgement about problematic inconsistency therefore requires converting the relative to an absolute effect.

The hypothetical example in Figure 3 in the body of the paper addresses this issue. We present how authors might have proceeded. Let us say the outcome in Figure 3 is mortality and the MID in mortality is 10 in 1,000 or 1%. Let us further say that the baseline risk of dying is 7%. Authors will then ask the question of what relative risk reduction corresponds to the 1% absolute MID.

As it turns out, the answer to that question is a 15% relative risk reduction, as illustrated in the following:

The formula for calculation the relative risk reduction that corresponds to the absolute MID is: Absolute MID / baseline risk

Using this formula in the above situation, the relative risk reduction corresponding to MID is as follows:

1% / 7% = 0.14 or 14% or approximately 15%.

The relative effect that corresponds to MID can be calculated as:

1 - relative risk reduction=1-0.15=0.85

Having decided, as described, on the relative effect that corresponds to MID one can include a vertical line – as in Figure 2 - vividly illustrating the relative effect that corresponds to the absolute MID.

Appendix 2. Instrument to assess the Credibility of Effect Modification Analyses (ICEMAN) in a meta-analysis of randomized controlled trials

Version 1.2

Consider the following important instructions informed by common misapplications of ICEMAN in studies using the instrument

- Complete a separate credibility assessment per each effect modifier (e.g., age, comorbidity, drug dose, etc.), outcome (e.g., mortality, stroke, duration of hospital stay), time-point (e.g., 3 months, 6 months), and effect measure (e.g. relative risk, risk difference).
- Do not apply ICEMAN if the interaction p-value is 0.1 or larger, i.e., provides very little statistical support for the existence of an effect modification (ICEMAN is designed to address the possible claim of an effect modification rather than the claim of no effect modification).
- Response options on the left indicate definitely or probably reduced credibility, response options on the right probably or definitely increased credibility
- Completely unclear should be interpreted as probably reduced credibility.
- To ensure transparency, provide a supporting comment under each question that provides a rationale for the rating.
- To ensure transparency, provide a copy of the completed ICEMAN instrument in the supplement of your article.

CREDIBILITY ASSESSMENT

Essential preliminary considerations to define the possible effect modification of interest

State a single outcome and time-point (e.g., mortality at 1 year follow-up): All-cause mortality

State a single effect measure (e.g., relative risk or risk difference): Relative risk

State a single candidate effect modifier (e.g., age or comorbidity): Magnitude of heart rate reduction

Was the effect modifier measured before or at randomization? [ ] yes, continue [**X**] no, refer to manual for further instructions

Comment: The effect modifier was measured after randomization in some studies. As discussed in the ICEMAN handbook, this does not invalidate an analysis based on a between-study comparison (like this one) and we can continue with the assessment.

1: Is the analysis of effect modification based on comparison within rather than between trials?

[**X**] Completely between	[ ] Mostly between or unclear	[ ] Mostly within	[ ] Completely within
Subgroup analysis or meta-regression comparing overall effects of each individual trial. This is typical for aggregate data meta-analysis.	*Subgroup analysis or meta-regression with most information coming from overall effects, but some trials providing within-trial subgroup information*	*Most trials providing within-trial subgroup information; or individual participant data analysis that combines within and between trial information*	*All trials providing within-trial subgroup information or individual participant data; and the analysis separates within from between trial information, e.g., meta-analysis of interactions*

Comment: The subgroup effect comes from a comparison between studies with higher and lower heart rate reductions on average.

2: For within-trial comparisons, is the effect modification similar from trial to trial? [**X**] Not applicable: no within-RCT comparison

[ ] Definitely not similar	[ ] Probably not similar or unclear	[ ] Mostly similar	[ ] Definitely similar
Effect modification reported for two or more trials and clearly different directions	*Effect modification not reported for individual trials or too imprecise to tell*	*Effect modification reported for two or more trials, mostly similar in direction, but considerable differences in magnitude*	*Effect modification reported for two or more trials, similar in direction, only some differences in magnitude*

Comment: NA

3: For between-trial comparisons, is the number of trials large? [ ] Not applicable: no between RCT comparison

[ ] Very small	[ ] Rather small or unclear	[ ] Rather large	[**X**] Large
1 or 2 or in smallest subgroup; 5 or less in continuous meta-regression	*3-4 in smallest subgroup; 6-10 in continuous meta-regression*	*5-9 in smallest subgroup; 11 to 15 in continuous meta-regression*	*10 or more in smallest subgroup; more than 15 in continuous meta-regression*

Comment: The authors included 23 studies and analyzed heart rate reductions as a continuous variable using meta-regression.

4: Was the direction of effect modification correctly hypothesized a priori?

[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[**X**] Definitely yes
Clearly post-hoc or results inconsistent with hypothesized direction or biologically very implausible	*Vague hypothesis or hypothesized direction unclear*	*No prior protocol available but unequivocal statement of a priori hypothesis with correct direction of effect modification*	*Prior protocol available and includes correct specification of direction of effect modification, e.g., based on a biologic rationale*

Comment: No prior protocol available but communication with authors established that this was a priori hypothesis with correct direction.

5: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification? (consider irrespective of number of effect modifiers)

[ ] Chance a very likely explanation	[ ] Chance a likely explanation or unclear	[X] Chance may not explain	[ ] Chance an unlikely explanation
Interaction or meta-regression p-value >0.05	*Interaction or meta-regression p-value ≤0.05 and >0.01, or no test of interaction reported and not computable*	*Interaction or meta-regression p-value ≤0.01 and >0.005*	*Interaction or meta-regression p-value ≤0.005*

Comment: P-value for test of interaction (in this case coming from meta-regression) is 0.006

6: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?

[X] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Explicitly exploratory analysis or large number of effect modifiers tested (e.g., greater than 10) and multiplicity not considered in analysis	*No mention of number or 4-10 effect modifiers tested and number not considered in analysis*	*No protocol available but unequivocal statement of 3 or fewer effect modifiers tested*	*Protocol available and 3 or fewer effect modifiers tested or number considered in analysis*

Comment: The authors tested 12 subgroup hypotheses

7: Did the authors use a random effects model?

[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[X] Definitely yes
Fixed (or common) effect or fixed effects model explicitly stated	*Probably fixed effect(s) model*	*Probably random (or mixed) effects*	*Random (or mixed) effects explicitly stated*

Comment: Communication with authors established that random effects meta-regression was used

8: If the effect modifier is a continuous variable, were arbitrary cut points avoided? [ ] not applicable: not continuous

[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[X] Definitely yes
Analysis based on exploratory cut point(s), e.g., picking cut point associated with highest interaction p-value	*Analysis based on cut point(s) of unclear origin*	*Analysis based on pre-specified cut point(s), e.g., suggested by prior RCT*	*Analysis based on the full continuum, e.g., assuming a linear or logarithmic relationship*

Comment: Heart rate reduction was treated as a continuous variable

9 Optional: Are there any additional considerations that may increase or decrease credibility? (manual section 3.9) [ ] not applicable

[ ] Yes, probably decrease [X] Yes, probably increase

Comment: Randomized trials of ivabradine in heart failure, another agent than beta-blocker that slows heart rate, have shown important benefit with heart rate slowing https://pubmed.ncbi.nlm.nih.gov/20801500/

10: How would you rate the overall credibility of the proposed effect modification?

The overall rating should be driven by the items that decrease credibility. The following provides a sensible strategy:

- All responses definitely or probably decrease credibility or unclear → very low
- Two or more responses definitely decrease credibility → maximum usually low even if all other responses satisfy credibility criteria
- One response definitely decreases credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- Two responses probably decrease credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- No response options definitely or probably decrease credibility → high very likely

Place a mark on the continuous line (or type "x" in editable version)

X

Very low credibility	**Low credibility**	**Moderate credibility**	**High credibility**
Minimal to no support for effect modification; Use overall effect for each subgroup	Some but insufficient support for effect modification; Use overall effect for each subgroup but note remaining uncertainty	Likely effect modification; Use separate effects for each subgroup but note remaining uncertainty	Very likely effect modification; Use separate effects for each subgroup

Comment: The most important limitations is the between-trial design and the large number of effect modifiers tested. Due to the large number of studies, the small p-value, and additional external evidence we rated the effect modification as moderate.